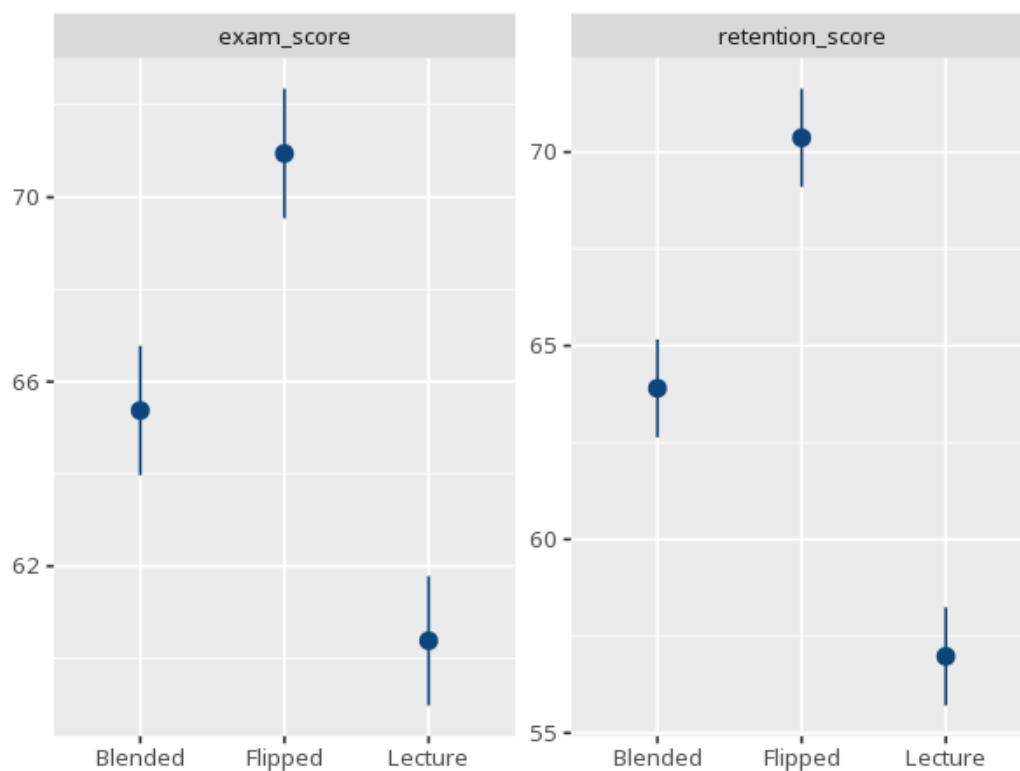


1 MANCOVA

Effect	Pillai / Wilks	approx F	df1	df2	p	partial η^2 [95% CI]
pretest_score	0.17	23.32	2	235	<.001	0.17 [0.10, 1.00]
teaching_method	0.48	37.47	4	472	<.001	0.24 [0.18, 1.00]

DV	F	df1	df2	p	partial η^2 [95% CI]
exam_score	55.07	2	236	<.001	0.32 [0.24, 1.00]
retention_score	108.57	2	236	<.001	0.48 [0.41, 1.00]



Like MANOVA, but group differences on the set of outcomes are assessed after controlling for one or more covariates. Interpret the univariate follow-ups only if the multivariate p is significant, and adjust them for the number of DVs (e.g. Bonferroni) to control familywise error. Your significance threshold (α) is 0.05.

A MANCOVA gave a covariate-adjusted group effect, Pillai's $V=.48$, $F(4,472)=37.47$, $p < .001$, partial $\eta^2=.24$ [.18, 1.00].

Statistical basis: Pillai's trace multivariate test (headline statistic, covariate-adjusted) (Pillai, K. C. S. (1955). "Some new test criteria in multivariate analysis." *Annals of Mathematical Statistics*, 26(1), 117-121.); Wilks' Lambda multivariate test (covariate-adjusted) (Wilks, S. S. (1932). "Certain generalized

distributions in multivariate analysis." *Biometrika*, 24(3/4), 471-494.); Box's M test of homogeneity of covariance matrices (Box, G. E. P. (1949). "A general distribution theory for a class of likelihood criteria." *Biometrika*, 36(3-4), 317-346.); Multivariate effect size (partial η^2 from the multivariate approx. F) (Ben-Shachar MS, Lüdtke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." *Journal of Open Source Software*, 5(56), 2815.); Estimated marginal (covariate-adjusted) means (Lenth RV (2024). *emmeans: Estimated Marginal Means, aka Least-Squares Means*. R package.).